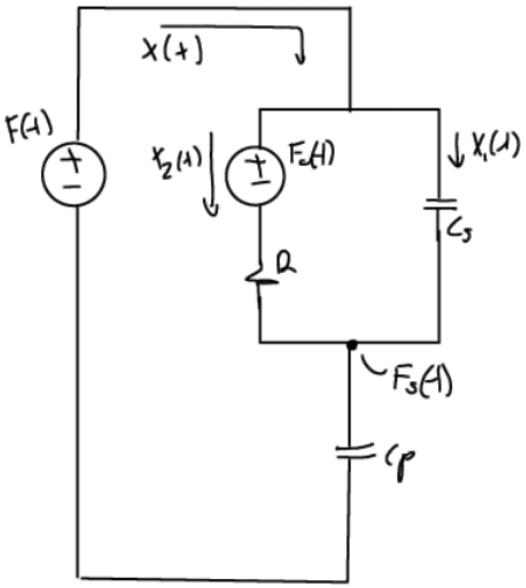
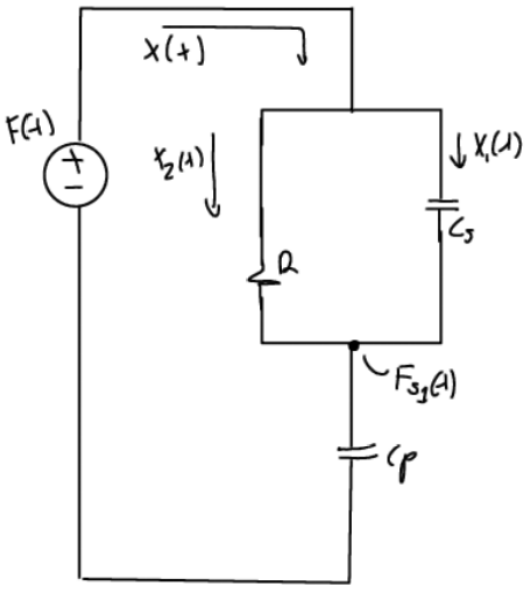
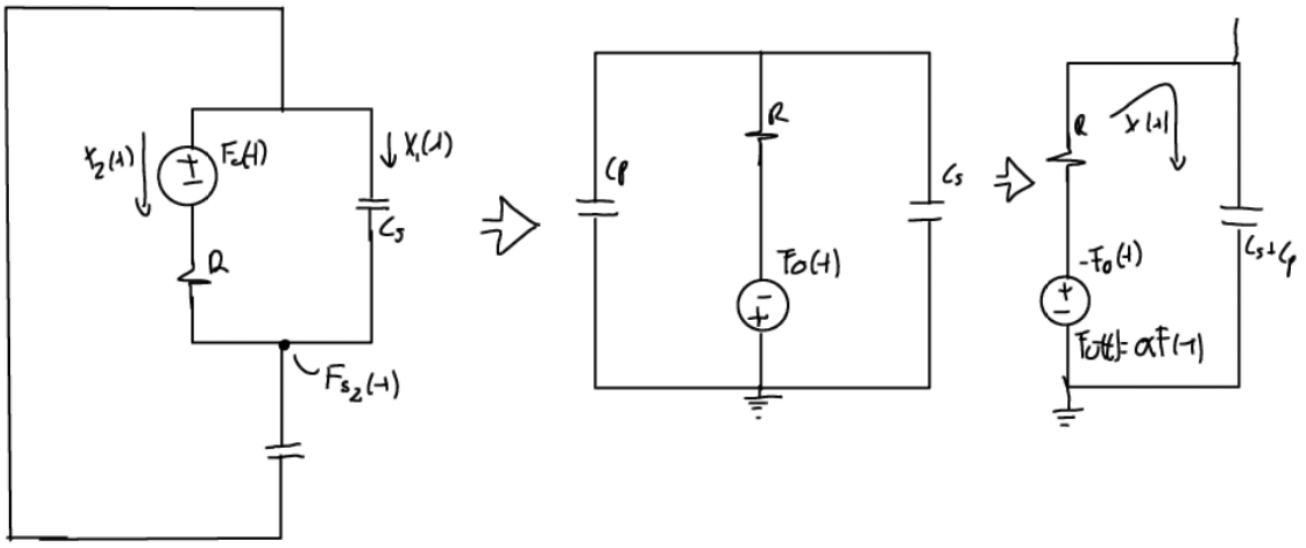
Circuito completo



Sin fuente $F_0(-1)$



Sin $F(-1)$



-Analizando circuito sin $F_0(t)$-

$$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{d[F_{s1}(t)]}{dt}$$

$X(t)$ = $C_p \frac{d[F_{s1}(t)]}{dt}$; $X_1(t)$ = $C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$; $X_2(t)$ = $\frac{F(t) - F_{s1}(t)}{R}$

$$C_p \frac{d[F_{s1}(t)]}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

Aplicando Laplace

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left[C_p s + C_s s + \frac{1}{R}\right] F_{s1}(s) = \left[C_s s + \frac{1}{R}\right] F(s)$$

$$\frac{R[C_p + C_s]s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{R[C_p + C_s]s + 1}{R}}{\frac{C_s R s + 1}{R} F(s)} = \frac{C_s R s + 1}{R[C_p + C_s]s + 1} F(s)$$

→ Salida 1

-Analizando circuito sin $F(t)$-

$$-\alpha F(t) = R x(t) + \frac{1}{C_s + C_p} \int x(t) dt$$

$$F_{s2}(t) = \frac{1}{C_s + C_p} \int x(t) dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{[C_p + C_s]s}$$

$$F_{s2}(s) = \frac{X(s)}{[C_p + C_s]s}$$

$$F(s) = -\frac{X(s)}{\alpha} \frac{R[C_p + C_s]s + 1}{[C_p + C_s]s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{[C_p + C_s]s}}{-\frac{X(s)}{\alpha} \frac{R[C_p + C_s]s + 1}{[C_p + C_s]s}} = -\frac{\alpha}{R[C_p + C_s]s + 1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R[C_p + C_s]s + 1}$$

→ Salida 2

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R[C_p + C_s]s + 1} F(s) - \frac{\alpha F(s)}{R[C_p + C_s]s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R[C_p + C_s]s + 1} F(s)$$

→ Salida Final